

ResUp

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Tanvi

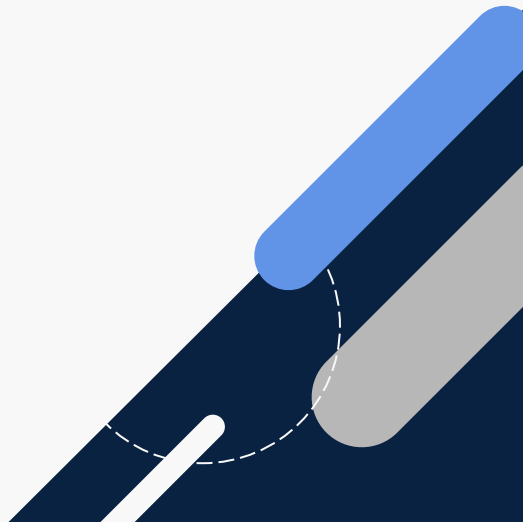




Introduction



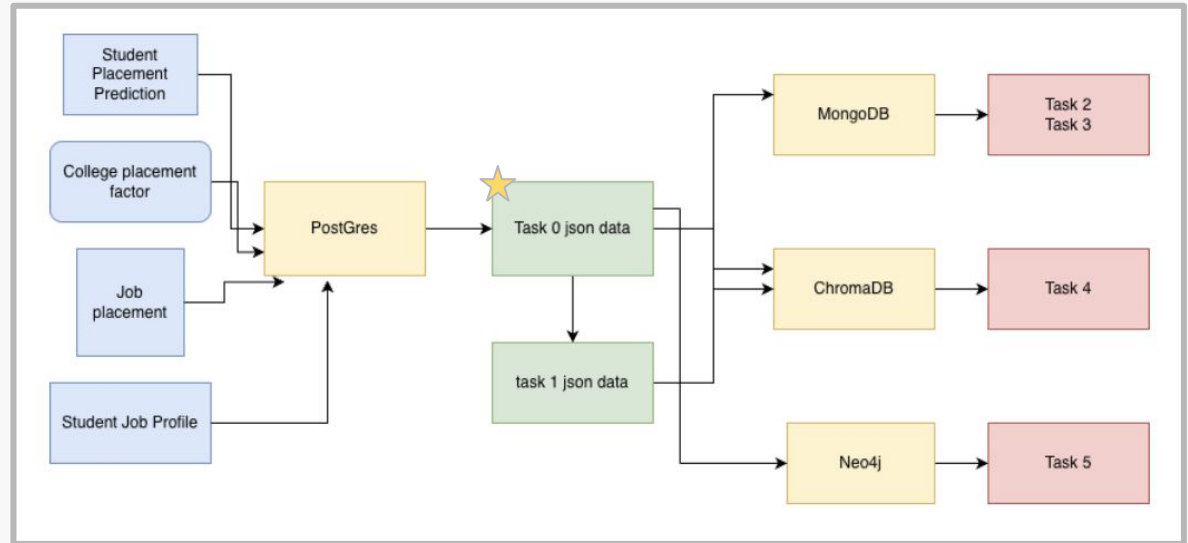
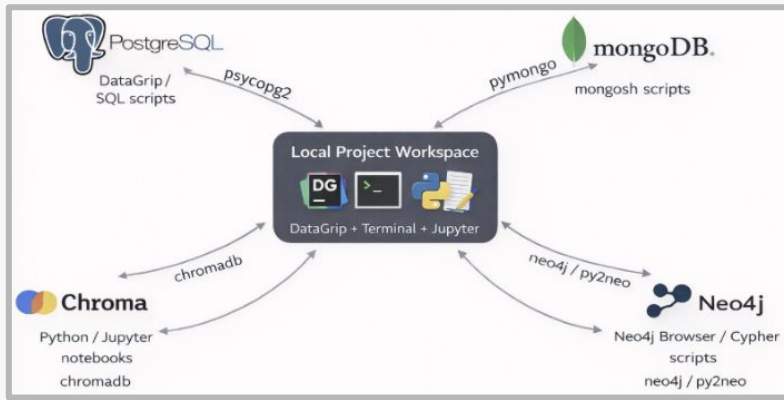
 Resume Data
 data
 masterResume
 task0
 task1
 task2
 task3
 task4
 task5





The Datasets & Databases



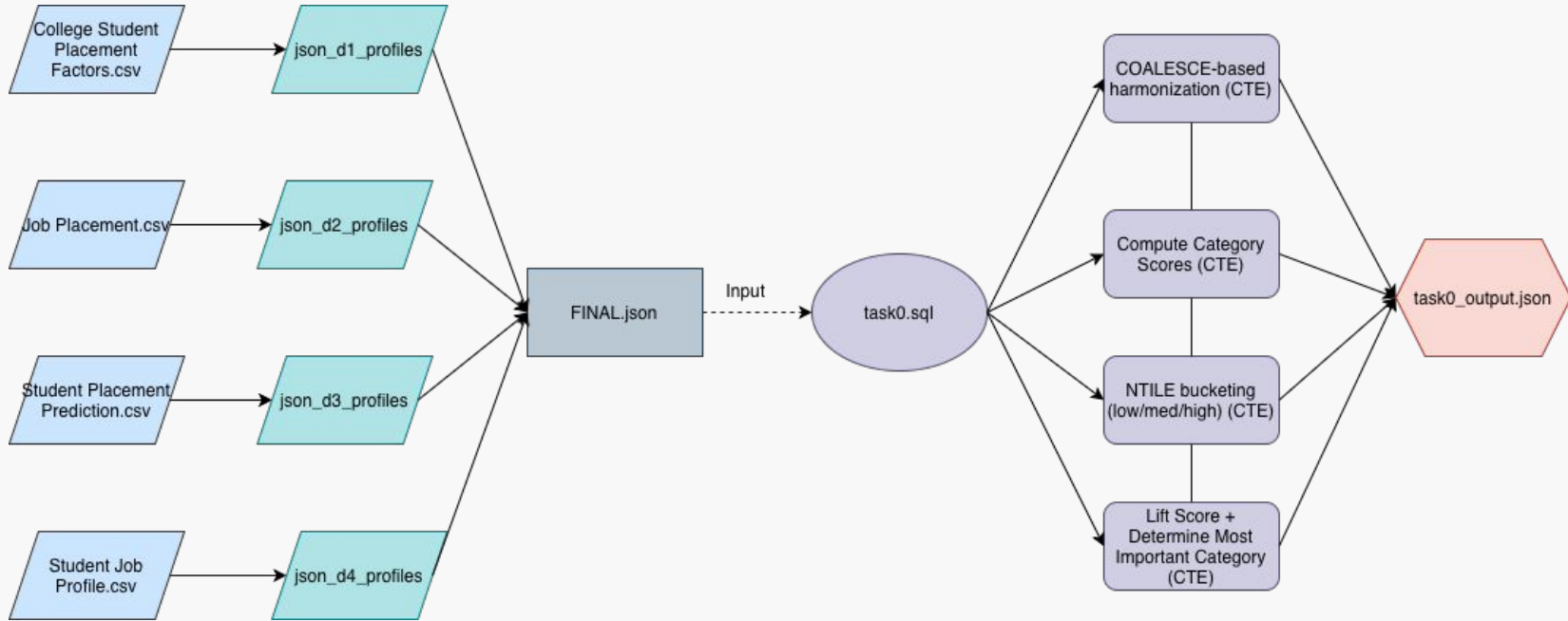




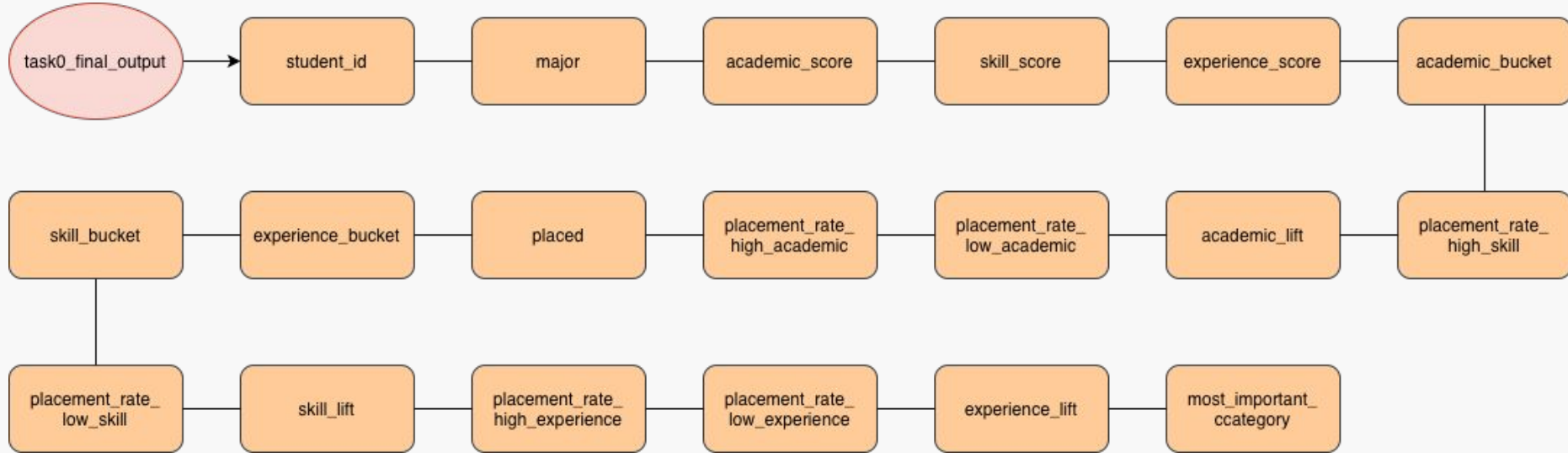
Task 0: Building the Database



Foundational Data Process Pipeline



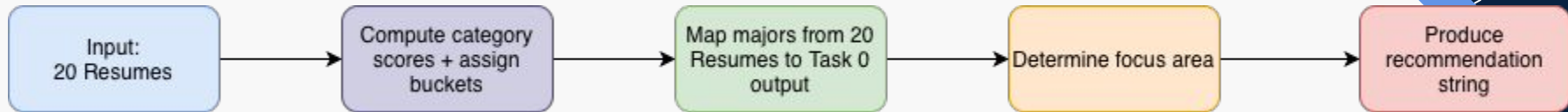
Baseline Data for Application





Task 1: Your Resume Performance





Step 1 & 2 repeat process described in Task 0.

Step 3

Determine focus area:

- ❑ Sorted by major
- ❑ Prioritize academics if CGPA is too low
 - ❑ A 3.0 GPA is usually a hard filter for most employers!
- ❑ Category with highest lift score is determined as the most important area
 - ❑ **Experience → Skill**
→ Academics



Step 4

Produce recommendation string

- ❑ URGENT
 - ❑ Focus category bucket is low, needs immediate attention
- ❑ GROW
 - ❑ Focus category bucket is currently medium
- ❑ MAINTAIN
 - ❑ Focus category bucket is already high

Sample Output

student_key	college	major	cgpa	academic_score	skill_score	experience_score	academic_bucket	skill_bucket	experience_bucket	most_important_category	focus_area
u_0016	UC Davis	Data Science	3.33	6.95	5.1	3.23	Medium	Medium	Medium	Experience	Experience
u_0017	UC Davis	Computer Science	2.94	6.36	5.1	3.23	Low	High	High	Experience	Academics
u_0018	UC Merced	Computer Science	3.09	6.59	5.1	3.23	Low	High	High	Experience	Experience

recommendation

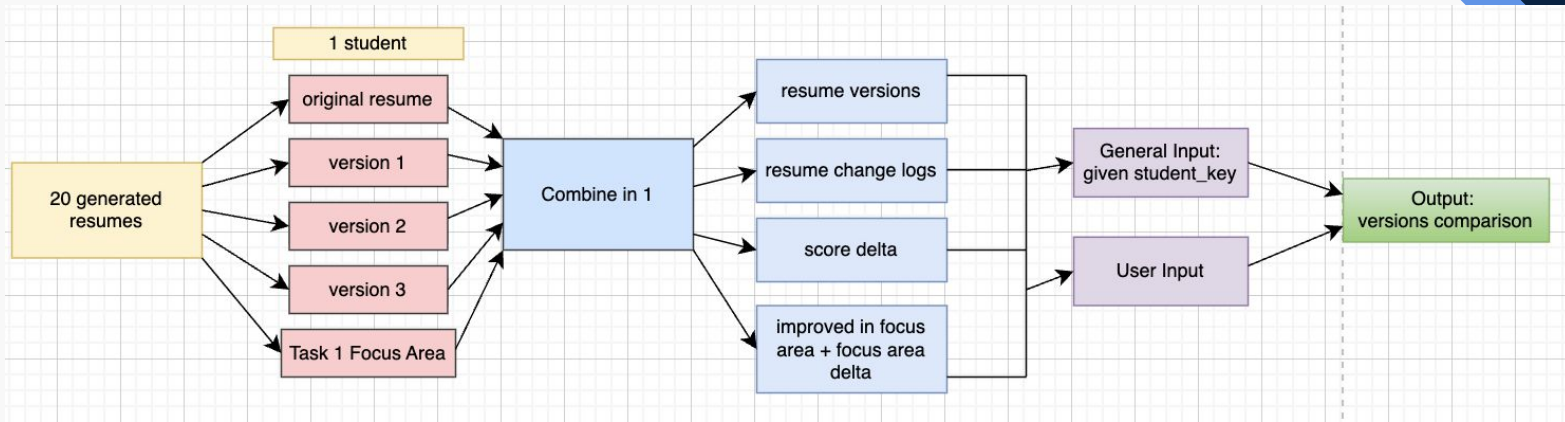
GROW — Push Experience from Medium to High. This is the highest-leverage move for Data Science placement (lift: 0.181). Take on a leadership role and add one more substantial project.
URGENT — GPA is 2.94 (below 3.0) and Academic score is Low. Raise your GPA first — this is a hard filter for most employers. Attend office hours, retake low-grade courses, and reduce extracurriculars temporarily.
MAINTAIN — Experience is already High. Keep it strong and shift focus to your next weakest area to stay well-rounded for Computer Science roles.





Task 2: Resume Tracking





STEP 1 : Merge resume versions

STEP 3 : Compare versions

STEP 2 : Score each version

```

db.resume_versions.find().forEach(doc => {
  db.resume_versions.updateOne(
    { _id: doc._id },
    {
      $set: {
        academic_score: academicScore(doc),
        skill_score: skillScore(doc),
        experience_score: experienceScore(doc)
      }
    }
  );
});

```

STEP 4 : Tie back to Task 1

- ❖ Uses Task 1's recommendation to check whether the student improved in the category that matters most



RESULT :

version to version

```
{
  student_key: 'u_0001',
  comparison_type: 'step',
  compared_versions: { from: 'v2', to: 'v3' },
  focus_area: null,
  score_before: { academic: 7.715, skill: 8.35, experience: 4.025 },
  score_after: { academic: 8.29, skill: 8.35, experience: 4.825 },
  score_delta: { academic: 0.575, skill: 0, experience: 0.8 },
  improved_in_focus_area: null,
  focus_area_delta: null,
  changes: {
    academics: {
      cgpa_old: 3.51,
      cgpa_new: 3.56,
      cgpa_delta: 0.05,
      coursework: { added: [ 'applied analytics' ], removed: [ ] },
      awards: { added: [ ], removed: [ ] }
    },
    skills: {
      technical: { added: [ 'pandas' ], removed: [ ] },
      soft: { added: [ ], removed: [ ] }
    },
    experience: {
      projects: { added: [ 'data quality checklist' ], removed: [ ] }
    },
    work_titles: { added: [ ], removed: [ ] },
    activities: { added: [ ], removed: [ ] }
  }
}
```

overall - all versions

```
{
  student_key: 'u_0001',
  comparison_type: 'overall',
  compared_versions: { from: 'original', to: 'v3' },
  focus_area: null,
  score_before: { academic: 7.065, skill: 5.1, experience: 3.225 },
  score_after: { academic: 8.29, skill: 8.35, experience: 4.825 },
  score_delta: { academic: 1.225, skill: 3.25, experience: 1.6 },
  improved_in_focus_area: null,
  focus_area_delta: null,
  changes: {
    academics: {
      cgpa_old: 3.41,
      cgpa_new: 3.56,
      cgpa_delta: 0.15,
      coursework: { added: [ 'project management', 'applied analytics' ], removed: [ ] },
      awards: { added: [ ], removed: [ ] }
    },
    skills: {
      technical: {
        added: [ 'unit testing', 'power bi', 'ci/cd', 'pandas' ],
        removed: [ ]
      },
      soft: { added: [ 'problem solving' ], removed: [ ] }
    },
    experience: {
      projects: {
        added: [ 'resume change log script', 'data quality checklist' ],
        removed: [ ]
      },
      work_titles: { added: [ ], removed: [ ] },
      activities: { added: [ ], removed: [ ] }
    }
  }
}
```

delta

added/
removed/
modified

Additional Feature! – the user can input which versions they want to compare

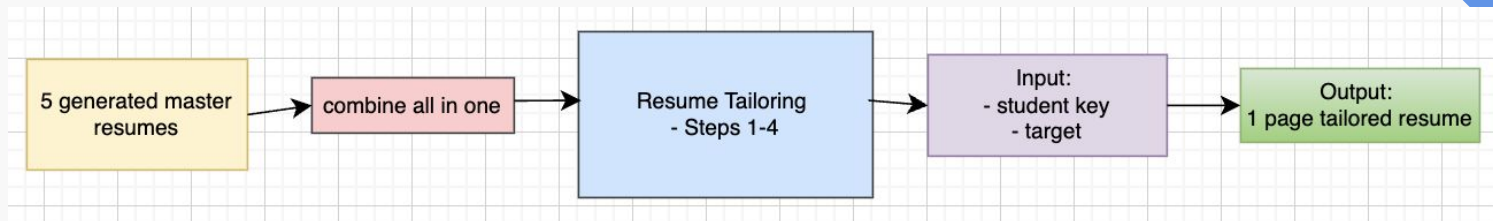


Task 3 : Tailored Resume



mongoDB

Special thanks to Jake!



STEP 1 : Resume → Bullets

STEP 2 : Score each bullet

```

- $addFields: {
  bullet_text: "$all_items.bullets",
  targetHits: { $size: { $regexFindAll: { input: "$combined_text", regex: targetRe } } },
  task0Hits: { $size: { $regexFindAll: { input: "$combined_text", regex: task0Re } } }
},
{ $match: { targetHits: { $gt: 0 } } },
{ $addFields: { score: { $add: [{ $multiply: ["$targetHits", 2] }, "$task0Hits"] } } },
{ $sort: { score: -1 } },

```

- ❖ Build a combined text field for each bullet
- ❖ Count target keyword matches [targetHits] and placement keyword matches [task0Hits]
- ❖ Score bullets (2*targetHits) + task0Hits



STEP 3 : Regroup & Trim

STEP 4 : Tailored Resume

```

$group: {
  _id: {
    section: "$all_items.section",
    title: "$all_items.title",
    org: "$all_items.org",
    start_date: "$all_items.start_date",
    end_date: "$all_items.end_date"
  },
  bullets: { $push: "$bullet_text" },
  bestScore: { $max: "$score" },
  education: { $first: "$education" },
  skills: { $first: "$skills" }
},
{
  $project: {
    _id: 0,
    section: "$_id.section",
    title: "$_id.title",
    org: "$_id.org",
    start_date: "$_id.start_date",
    end_date: "$_id.end_date",
    bullets: { $slice: ["$bullets", bulletsPerItem] },
    education: 1,
    skills: 1,
    bestScore: 1
  },
  $sort: { bestScore: -1 },
  $limit: maxItemsTotal,
}

```


RESULT :

Input : runTailor("u_0001", "data analyst internship sql python healthcare")

```
{
  student key: 'u_0001',
  target: 'data analyst internship sql python healthcare',
  resume: {
    education: [
      {
        institution: 'University of California, Davis',
        degree: 'M.S.',
        field_of_study: 'Data Science',
        start_date: '2024-09',
        end_date: '2026-06',
        status: 'in_progress',
        cgpa: 3.82,
        coursework: [
          {
            code: 'DSC 241',
            name: 'Statistical Models',
            topics: [
              'multiple regression',
              'diagnostics',
              'model selection (AIC/CV)',
              'bootstrap',
              'generalized linear models',
              'penalized regression',
              'nonparametric regression'
            ]
          },
          {
            code: 'ECS 171',
            name: 'Machine Learning'
          },
          {
            code: 'STA 207',
            name: 'Applied Regression'
          },
          {
            code: 'ECS 272',
            name: 'Information Visualization'
          }
        ]
      },
      {
        code: 'STA 207',
        name: 'Applied Regression'
      },
      {
        code: 'ECS 272',
        name: 'Information Visualization'
      }
    ],
    highlights: [
      'Teaching support role (2025): led lab sections/office hours and helped grade assignments.'
    ]
  }
}
```

```
skills: {
  programming_languages: [
    'Python',
    'SQL',
    'R',
    'JavaScript',
    'TypeScript'
  ],
  ml_data: [
    'pandas',
    'numpy',
    'scikit-learn',
    'statsmodels',
    'xgboost (basic)',
    'PyTorch (basic)'
  ],
  visualization: [
    'matplotlib',
    'plotly',
    'Tableau',
    'Power BI'
  ],
  data_engineering: [
    'ETL (Python/SQL)',
    'Airflow (basic)',
    'dbt (basic)'
  ],
  databases: [
    'PostgreSQL',
    'MySQL',
    'SQLite',
    'BigQuery (basic)'
  ],
  cloud_tools: [
    'AWS (S3, Lambda) (basic)',
    'Docker (basic)',
    'GitHub Actions'
  ],
  soft_skills: [
    'technical writing',
    'presentation',
    'stakeholder communication',
    'team collaboration'
  ]
}
```

```
experience: [
  {
    section: 'experience',
    title: 'Data Science Intern',
    org: 'Nimbus Health Analytics',
    start_date: '2025-06',
    end_date: '2025-09',
    bullets: [
      'Designed a feature pipeline in SQL + Python (120+ features) with automated data quality checks and reproducible notebooks.',
      'Built a claims-based patient risk model using logistic regression and gradient boosting; improved AUC from 0.71 to 0.79 on validation data.'
    ]
  },
  {
    section: 'experience',
    title: 'Student Developer',
    org: 'Campus IT Services (UCSD)',
    start_date: '2022-01',
    end_date: '2023-06',
    bullets: [
      'Built an internal ticket triage tool integrating email + database records; reduced manual tagging workload by ~25%.',
      'Wrote SQL queries and exports for monthly service metrics used in reporting.'
    ]
  }
],
projects: [
  {
    section: 'projects',
    title: 'Personal Finance Dashboard',
    org: 'personal project',
    start_date: '2023-08',
    end_date: '2023-12',
    bullets: [
      'Ingested transaction data and built automated category rules created interactive spending trend dashboards.'
    ]
  }
],
research: []
}
```

2 experiences

1 project

RESULT :

Input : runTailor("u_0001", "IT services")

Reduced to only 1 experience

```
{
  student_key: 'u_0001',
  target: 'IT services',
  resume: {
    education: [
      {
        institution: 'University of California, Davis',
        degree: 'M.S.',
        field_of_study: 'Data Science',
        start_date: '2024-09',
        end_date: '2026-06',
        status: 'in_progress',
        cgpa: 3.82,
        coursework: [
          {
            code: 'DSC 241',
            name: 'Statistical Models',
            topics: [
              'multiple regression',
              'diagnostics',
              'model selection (AIC/CV)',
              'bootstrap',
              'generalized linear models',
              'penalized regression',
              'nonparametric regression'
            ]
          },
          {
            code: 'ECS 471',
            name: 'Machine Learning'
          },
          {
            code: 'STA 207',
            name: 'Applied Regression'
          },
          {
            code: 'ECS 272',
            name: 'Information Visualization'
          }
        ]
      },
      {
        code: 'ECS 171',
        name: 'Machine Learning'
      },
      {
        code: 'STA 207',
        name: 'Applied Regression'
      },
      {
        code: 'ECS 272',
        name: 'Information Visualization'
      }
    ],
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skills: {
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    'R',
    'JavaScript',
    'TypeScript'
  ],
  ml_data: [
    'pandas',
    'numpy',
    'scikit-learn',
    'statsmodels',
    'xgboost (basic)',
    'pytorch (basic)'
  ],
  visualization: [
    'matplotlib',
    'plotly',
    'Tableau',
    'Power BI'
  ],
  data_engineering: [
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    'Airflow (basic)',
    'dbt (basic)'
  ],
  databases: [
    'PostgreSQL',
    'MySQL',
    'SQLite',
    'BigQuery (basic)'
  ],
  cloud_tools: [
    'AWS (S3, Lambda) (basic)',
    'Docker (basic)',
    'GitHub Actions'
  ],
  soft_skills: [
    'technical writing',
    'presentation',
    'stakeholder communication',
    'team collaboration'
  ]
},
experience: [
  {
    section: 'experience',
    title: 'Student Developer',
    org: 'Campus IT Services (UCSD)',
    start_date: '2022-01',
    end_date: '2023-06',
    bullets: [
      'Built an internal ticket triage tool integrating email + database records; reduced manual tagging workload by ~25%',
      'Wrote SQL queries and exports for monthly service metrics used in reporting.'
    ]
  }
],
projects: [],
research: []
}
```

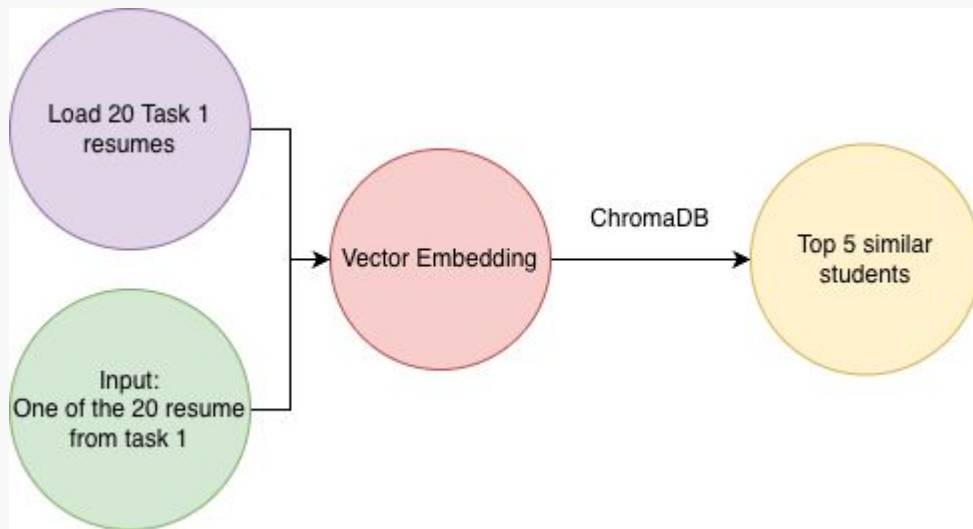


Task 4: Similarity and Placement Prediction



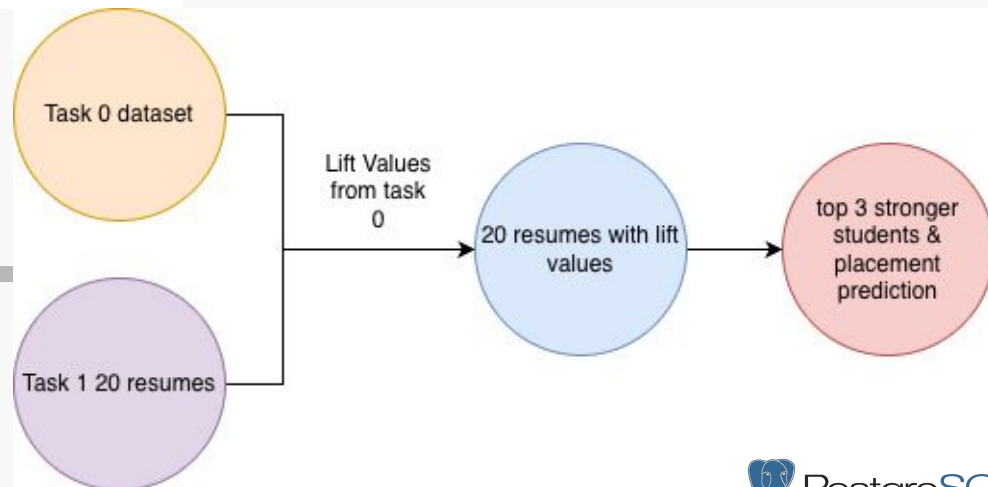
CHROMADB





Who has the most similarity to me?

Who has the best predicted placement odds?



$(\text{academic score} \times \text{academic lift}) + (\text{skill score} \times \text{skill lift}) + (\text{experience score} \times \text{experience lift})$

Student Key – u_0007 | Computer Science | CGPA: 3.62

Scores: acad=7.38 skill=5.1 exp=3.23

Buckets: High / Low / Low

placement_prediction: 28.9%

OUTPUT 1 – Top 5 Similar Resume Peers (collection rich)

Question: "Which of the 20 detailed resume peers is most similar to me?"

Method: Full resume text embedding (semantic similarity)

student_key	distance	major	cgpa	academic_score	skill_score	experience_score	placement_prediction
u_0020	0.0469	Computer Science	3.95	7.8800	6.1500	3.2300	30.6000%
u_0004	0.0542	Data Science	3.88	7.7700	5.1000	3.0500	35.3000%
u_0016	0.0628	Data Science	3.33	6.9500	5.1000	3.2300	30.8000%
u_0001	0.0711	Computer Science	3.41	7.0700	5.1000	3.2300	28.9000%
u_0013	0.0786	Computer Science	3.47	7.1600	5.1000	3.2300	28.9000%

OUTPUT 2 – Top 3 Stronger Comparison Students (PostgreSQL)

Question: "Who has better placement odds than me in CSE?"

NOTE: Queried across ALL 2,020 students – not filtered from Output 1.

Your placement percentile: top 93.7% (outperform 127 of 2020 students)

student_key	major	academic_score	skill_score	experience_score	placement_prediction
d3_469	CSE	8.3600	25.2503	5.3333	79.2%
d3_295	CSE	8.6850	25.7148	4.6667	74.0%
d3_1493	CSE	7.8853	23.8216	5.0000	74.0%



IMPROVEMENT DIRECTION (vs top stronger peer):

- ▲ Academic Score : you=36.5% | peer=60.3% | gap=+23.7%
- ▲ Skill Score : you=0.0% | peer=78.0% | gap=+78.0%
- ▲ Experience Score: you=48.4% | peer=80.0% | gap=+31.5%

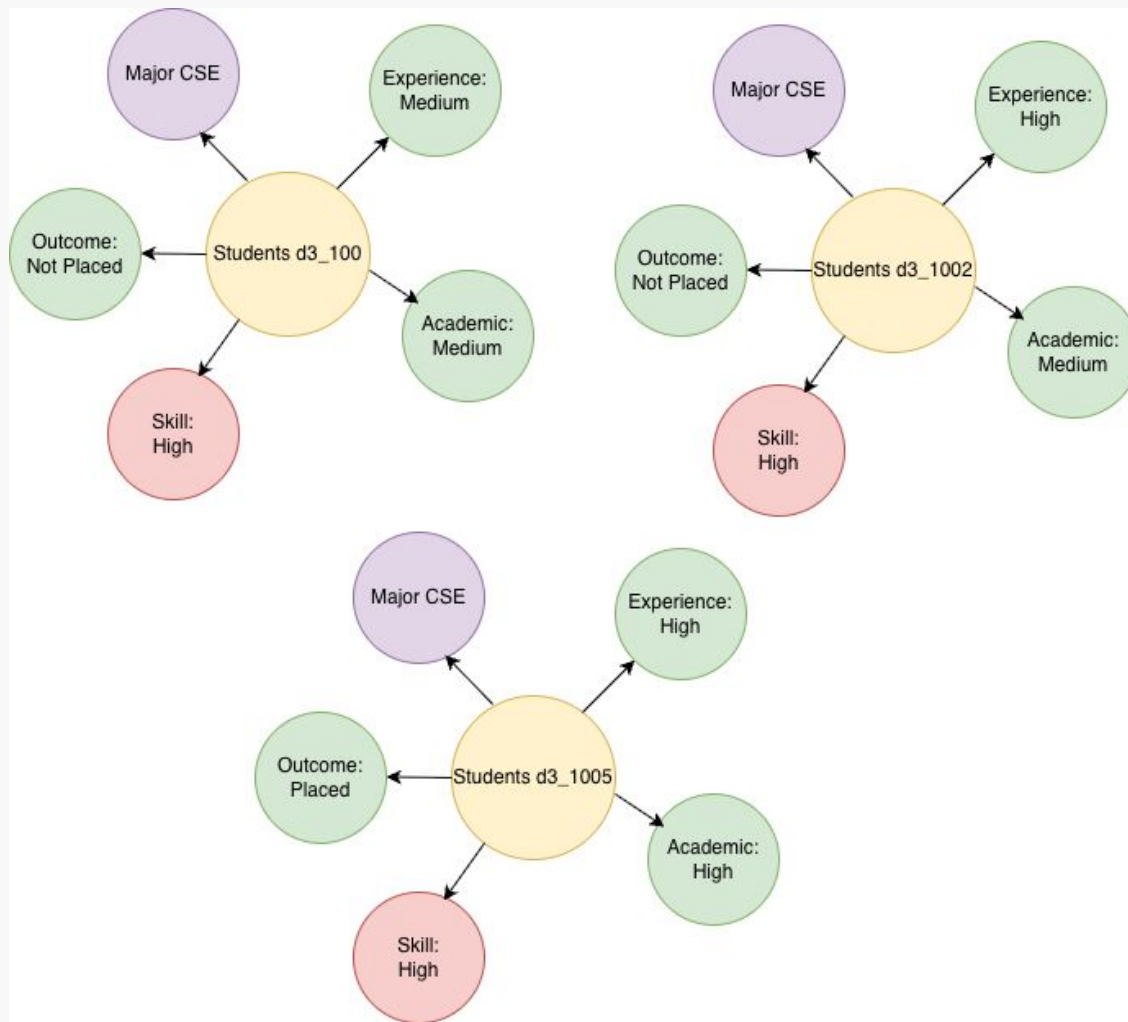


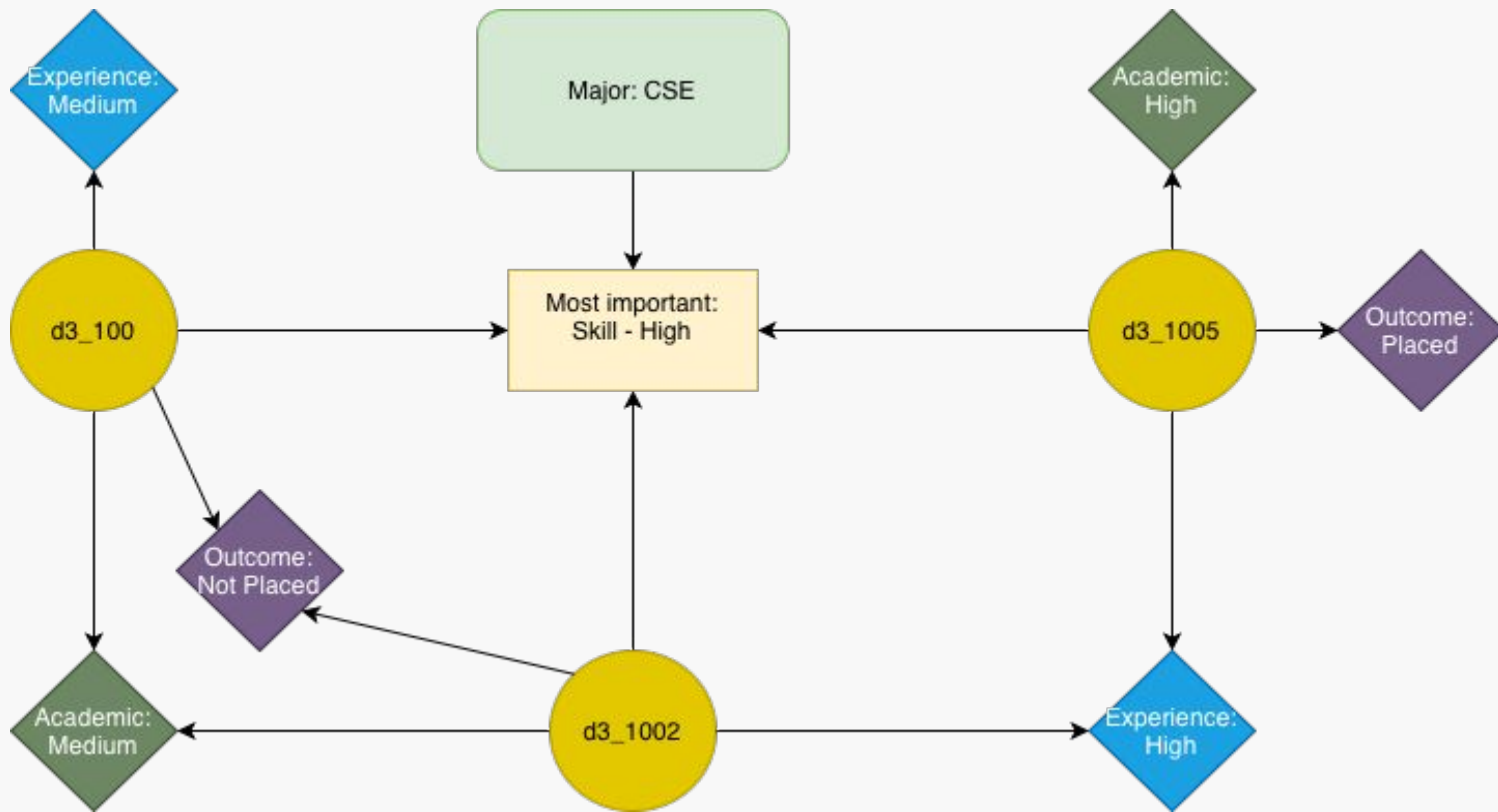
CHROMADB



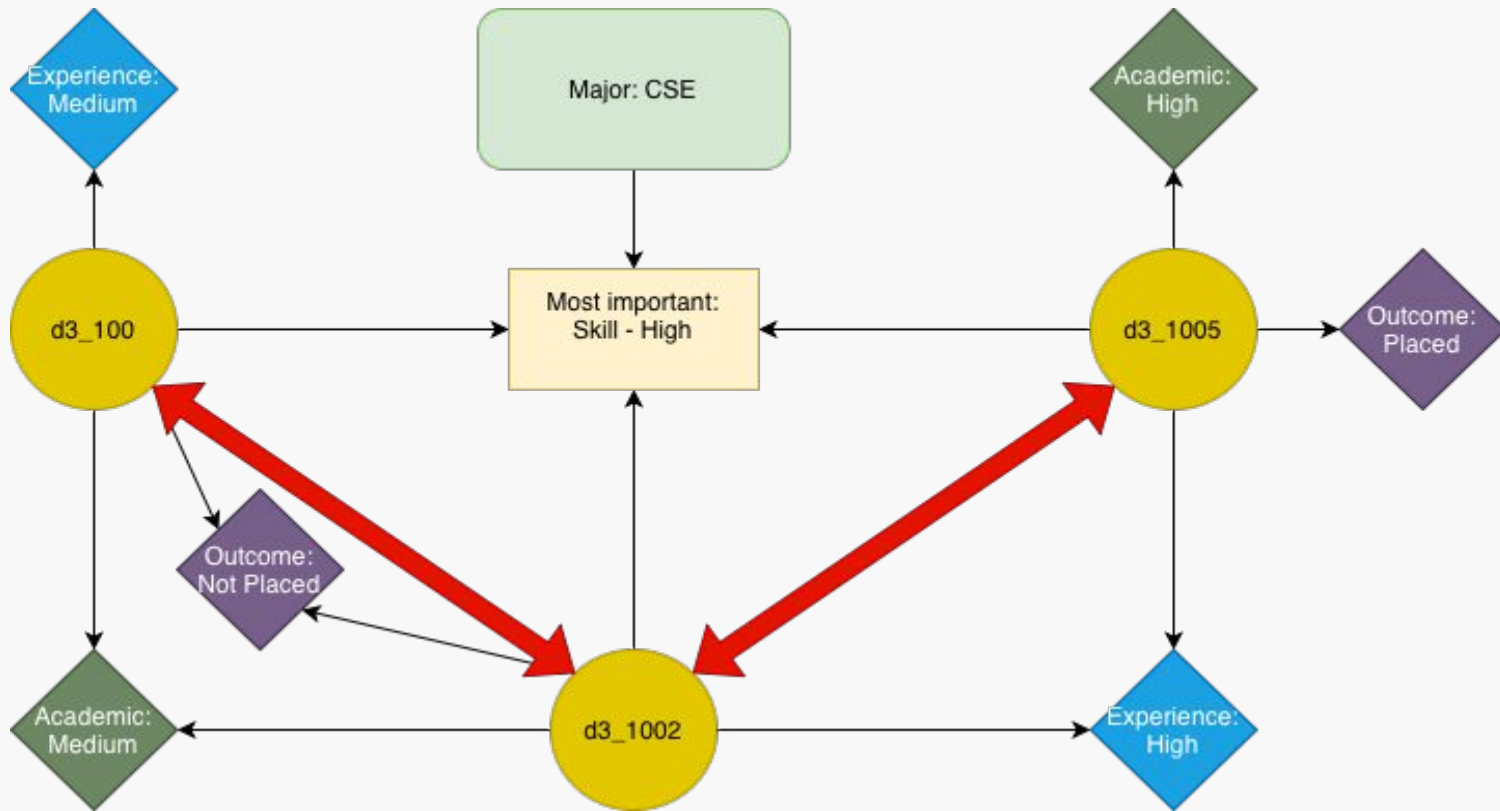
Task 5: Peer map







...



...

```
=====
STUDENT: d3_100 | Major: CSE | Status: NotPlaced
Profile: Academic=Medium | Skill=High | Experience=Medium
=====
```

LAYER 1 – GRAPH DISTANCE ANALYSIS:

```
-----
You are 1 hop(s) from the nearest placed CSE peer.
Nearest placed peer: d3_1035
5 placed CSE students within 1 hop(s).
```

LAYER 2 – PEER EVIDENCE (from graph):

```
-----
Among 404 placed CSE students:
→ 100% have Skill:High ✓ you have this
→ 60% have Academic:High ← MISSING
→ 53% have Experience:High ← MISSING

54 placed peers share your exact profile
(Acad:Medium, Skill:High, Exp:Medium) = 13% of placed CSE.
```

LAYER 3 – GRAPH RECOMMENDATION:

```
-----
Priority 1 – EXPERIENCE (lift: +0.107)
  Upgrading Experience: Medium → High
  connects you to +67 more placed CSE students.

Priority 2 – ACADEMIC (lift: -0.058)
  Lift is NEGATIVE for CSE – do NOT prioritize Academic.

Priority 3 – SKILL (lift: +0.072)
  Already optimal – maintain Skill:High.
```

```
-----
— BOTTOM LINE —
One change (Experience: Medium → High) connects you to +67 more placed peers in CSE.
```

```
=====
STUDENT: d3_1005 | Major: CSE | Status: Placed
Profile: Academic=High | Skill=High | Experience=High
=====
```

LAYER 1 – GRAPH DISTANCE ANALYSIS:

```
-----
You are 1 hop(s) from the nearest placed CSE peer.
Nearest placed peer: d3_1011
5 placed CSE students within 1 hop(s).
```

LAYER 2 – PEER EVIDENCE (from graph):

```
-----
Among 404 placed CSE students:
→ 100% have Skill:High ✓ you have this
→ 60% have Academic:High ✓ you have this
→ 53% have Experience:High ✓ you have this

129 placed peers share your exact profile
(Acad:High, Skill:High, Exp:High) = 32% of placed CSE.
```

LAYER 3 – GRAPH RECOMMENDATION:

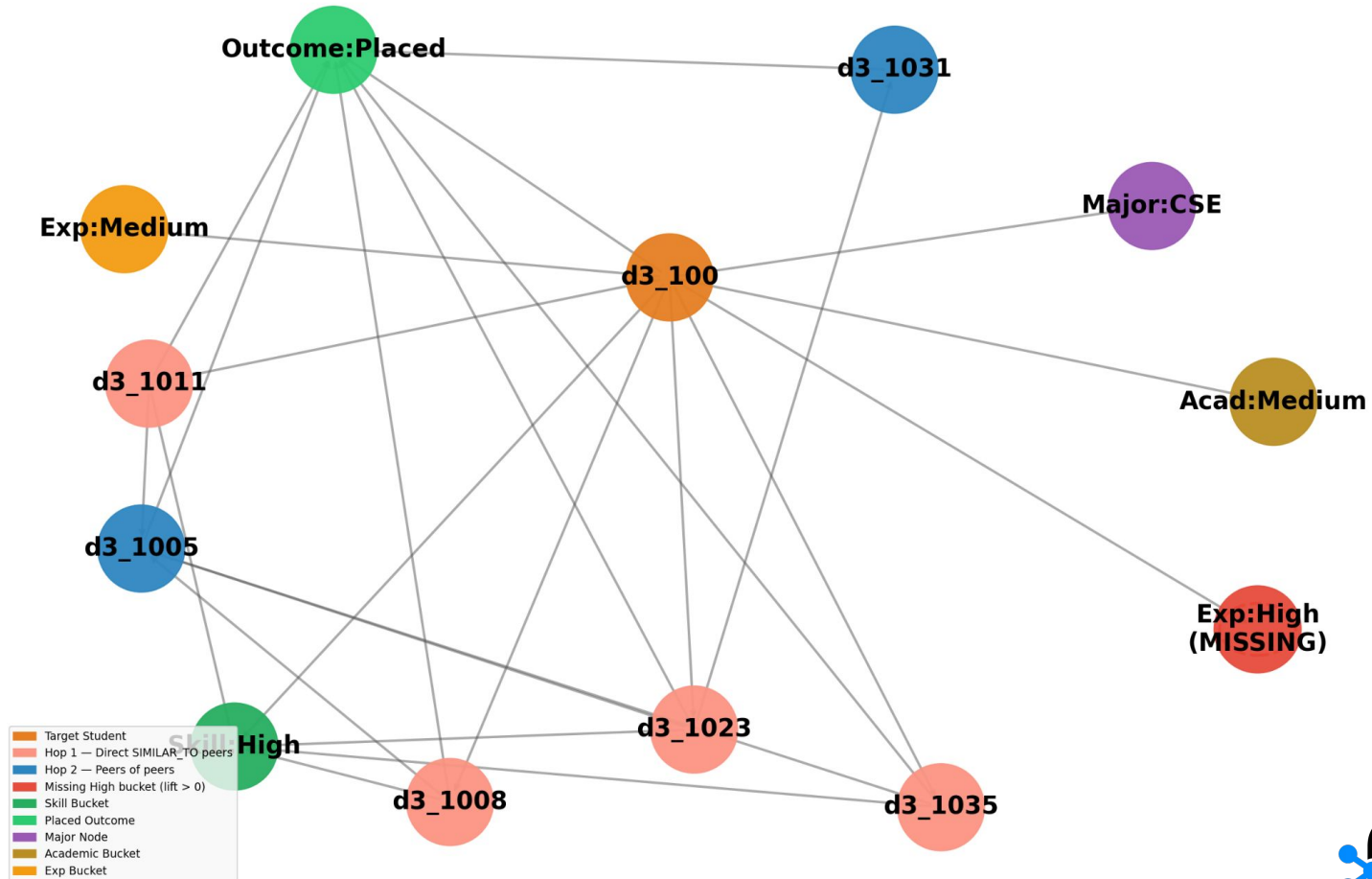
```
-----
Priority 1 – ACADEMIC (lift: -0.058)
  Already optimal – maintain Academic:High.

Priority 2 – SKILL (lift: +0.072)
  Already optimal – maintain Skill:High.

Priority 3 – EXPERIENCE (lift: +0.107)
  Already optimal – maintain Experience:High.
```

```
-----
— BOTTOM LINE —
No positive-lift upgrade is recommended for this profile.
```

Peer Placement Network — d3_100 (CSE)





**So, what are the next
steps?**



Limitations + Future Directions

Limitations

- ❑ Current task 0 data is fixed historical data
- ❑ Uses synthetic resumes
- ❑ Tasks are operating independently

Future Directions

- ❑ Replace current foundational data
 - ❑ Historical user data
 - ❑ More trusted sources
- ❑ Implement real resume ingestion
- ❑ Create pipeline connecting all tasks
- ❑ Create user interface





Thank You!

